Dimensionality Reduction and Feature Selection for Network Intrusion Detection on Apache Spark ML

…with a Leakage-Aware Evaluation Pipeline

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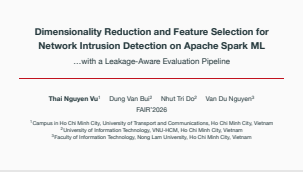


Seven minutes on a single claim: on CICIDS2017, a near-perfect F1 is usually a property of the evaluation pipeline, not of the model.

I. PROBLEM AND CONTRIBUTIONS

A near-perfect F1 on CICIDS2017 is a warning sign

Two problems, usually treated separately

- **Cost.** ≈80 features per flow.
- **Trust.** Near-perfect binary F1 usually signals **data leakage**.

Two concrete leakage sources

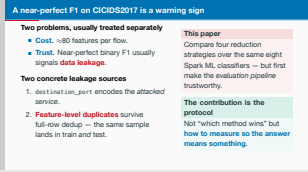
1. `destination_port` encodes the *attacked service*.
2. **Feature-level duplicates** survive full-row dedup — the same sample lands in train *and* test.

This paper

Compare four reduction strategies over the same eight Spark ML classifiers — but first make the *evaluation pipeline* trustworthy.

The contribution is the protocol

Not "which method wins" but **how to measure so the answer means something**.



Frame the talk as a measurement paper. The port gives the model a port-to-label shortcut; 80 features are expensive to train, to score and to fit on an edge device. Both leakage sources are removed before any split.

Contributions

1. A **uniform comparison** of RF Importance, SHAP and PCA across the same 8 Spark ML algorithms + 2 ensembles on CICIDS2017.
2. SHAP contrasted **directly** against RF Importance on the same training set.
3. A **leakage-aware, statistically tested** pipeline: leak removal, deterministic feature-level dedup, validation-based selection, exact paired permutation test with Nadeau–Bengio correction.
4. A **per-attack-type multiclass** evaluation — where the methods genuinely differ.
5. An **edge-aware** recommendation: F1 *and* feature count *and* model size, for a Jetson Orin Nano Super (deployment reported in a companion systems paper).

Contributions
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Five contributions; three, four and five are the ones rarely done together.

II. LEAKAGE-AWARE PIPELINE

The leakage-aware pipeline

CICIDS2017 — 8 CSV files, ≈2.8M flows

↓

Leakage-aware preparation
remove `destination_port` · feature-level dedup

↓

80/20 split (fixed seed) · Parquet

↓

4 strategies: Baseline | RF Top-*K* | SHAP Top-*K* | PCA
(rankings fit on the *training set only*)

↓

8 Spark ML classifiers + 2 ensembles

↓

Validation-based selection · paired permutation test
Nadeau–Bengio CI · effect size · multiclass

↓

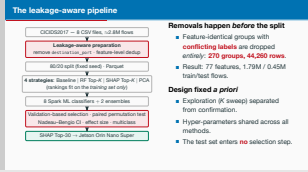
SHAP Top-30 → Jetson Orin Nano Super

Removals happen *before* the split

- Feature-identical groups with **conflicting labels** are dropped *entirely*: **270 groups, 44,260 rows**.
- Result: 77 features, 1.79M / 0.45M train/test flows.

Design fixed *a priori*

- Exploration (*K* sweep) separated from confirmation.
- Hyper-parameters shared across all methods.
- The test set enters **no** selection step.



Deterministic deduplication is the part people get wrong. Conflicting groups are removed wholesale, so the result does not depend on how Spark partitions the data.

III. RESULTS AND CONCLUSION

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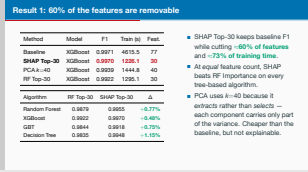
Result 1: 60% of the features are removable

Method	Model	F1	Train (s)	Feat.
Baseline	XGBoost	0.9971	4615.5	77
SHAP Top-30	XGBoost	**0.9970**	**1226.1**	**30**
PCA k=40	XGBoost	0.9939	1444.8	40
RF Top-30	XGBoost	0.9922	1295.1	30

Algorithm	RF Top-30	SHAP Top-30	Δ
Random Forest	0.9879	0.9955	**+0.77%**
XGBoost	0.9922	0.9970	**+0.48%**
GBT	0.9844	0.9918	**+0.75%**
Decision Tree	0.9835	0.9948	**+1.15%**

- SHAP Top-30 keeps baseline F1 while cutting ≈**60% of features** and ≈**73% of training time**.
- At *equal* feature count, SHAP beats RF Importance on every tree-based algorithm.
- PCA uses k=40 because it *extracts* rather than *selects* — each component carries only part of the variance. Cheaper than the baseline, but not explainable.



Two tables, one message: SHAP Top-30 is the operating point. Same accuracy, 60 percent fewer features, and it is explainable, which matters for the SOC analyst and for the edge board downstream.

Result 2: testing the difference instead of asserting it

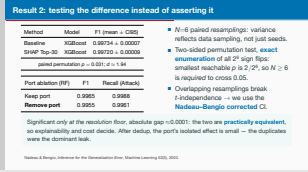
Method	Model	F1 (mean ± CI95)
Baseline	XGBoost	0.99734 ± 0.00007
SHAP Top-30	XGBoost	0.99720 ± 0.00009
paired permutation $p = 0.031$, $d \approx 1.94$		

Port ablation (RF)	F1	Recall (Attack)
Keep port	0.9965	0.9988
Remove port	0.9955	0.9961

- N=6 paired *resamplings*: variance reflects data sampling, not just seeds.
- Two-sided permutation test, **exact enumeration** of all 2^6 sign flips: smallest reachable p is $2/2^6$, so $N \geq 6$ is *required* to cross 0.05.
- Overlapping resamplings break t-independence $\rightarrow$ we use the **Nadeau–Bengio corrected** CI.

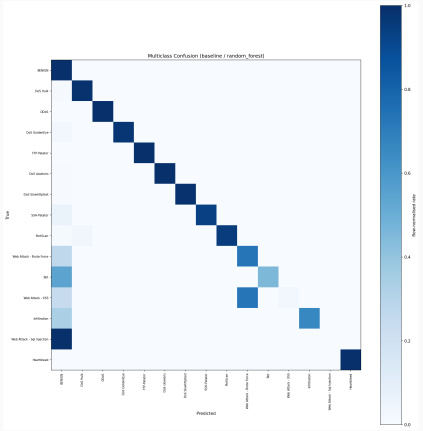
Significant *only at the resolution floor*, absolute gap $\approx$0.0001: the two are **practically equivalent**, so explainability and cost decide. After dedup, the port's isolated effect is small — the duplicates were the dominant leak.

Nadeau & Bengio, *Inference for the Generalization Error*, Machine Learning 52(3), 2003.



Variance reflects data sampling, not just model initialization. The test is two-sided, and overlapping resamplings break t-independence. Small p, tiny effect in absolute terms: say plainly that we treat them as equivalent rather than claiming a winner.

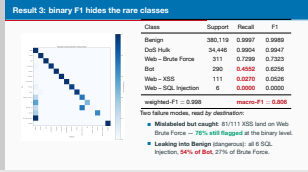
Result 3: binary F1 hides the rare classes



Class	Support	Recall	F1
Benign	380,119	0.9997	0.9989
DoS Hulk	34,446	0.9904	0.9947
Web – Brute Force	311	0.7299	0.7323
Bot	290	**0.4552**	0.6256
Web – XSS	111	**0.0270**	0.0526
Web – SQL Injection	6	**0.0000**	0.0000
weighted-F1 = 0.998		**macro-F1 = 0.806**	

Two failure modes, *read by destination*:

- **Mislabeled but caught**: 81/111 XSS land on Web Brute Force — **76% still flagged** at the binary level.
- **Leaking into Benign** (dangerous): all 6 SQL Injection, **54% of Bot**, 27% of Brute Force.



Weighted F1 0.998 versus macro F1 0.806. Three causes: extreme imbalance; removing the port hits web attacks hardest, because without the port-80 shortcut short HTTP flows look like browsing, so the 0.027 recall reported here replaces a leakage-inflated one; and Bot C&C beaconing resembles background traffic.

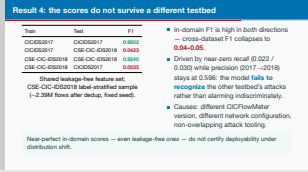
Result 4: the scores do not survive a different testbed

Train	Test	F1
CICIDS2017	CICIDS2017	0.9952
CICIDS2017	CSE-CIC-IDS2018	0.0423
CSE-CIC-IDS2018	CSE-CIC-IDS2018	0.9245
CSE-CIC-IDS2018	CICIDS2017	0.0535

Shared leakage-free feature set; CSE-CIC-IDS2018 label-stratified sample (~2.39M flows after dedup, fixed seed).

- In-domain F1 is high in *both* directions — cross-dataset F1 collapses to **0.04–0.05**.
- Driven by near-zero *recall* (0.022 / 0.030) while precision (2017→2018) stays at 0.596: the model **fails to recognize** the other testbed's attacks rather than alarming indiscriminately.
- Causes: different CICFlowMeter version, different network configuration, non-overlapping attack tooling.

Near-perfect in-domain scores — *even leakage-free ones* — do not certify deployability under distribution shift.



Do not soften this. In-domain 0.99, cross-dataset 0.04. Consistent with prior cross-dataset IDS studies, and the reason domain adaptation is a prerequisite rather than a nice-to-have.

Conclusion

What we showed

- SHAP Top-30 retains baseline F1 with **≈60% fewer features**.
- Baseline vs. SHAP Top-30 are **practically equivalent**.
- **macro-F1 0.806** against weighted-F1 0.998.
- Cross-dataset F1 **collapses to 0.04–0.05**.

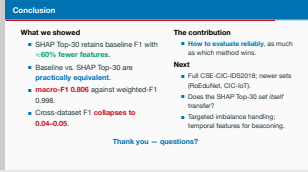
The contribution

- **How to evaluate reliably**, as much as which method wins.

Next

- Full CSE-CIC-IDS2018; newer sets (RoEduNet, CIC-IoT).
- Does the SHAP Top-30 *set itself* transfer?
- Targeted imbalance handling; temporal features for beaconing.

Thank you — questions?



Close on the protocol, not the leaderboard.

Key references

[1] Sharafaldin, I., Lashkari, A. H. & Ghorbani, A. A. *Toward generating a new intrusion detection dataset and intrusion traffic characterization*. ICISSP 2018. **CICIDS2017**

[2] Lanvin, M. et al. *Errors in the CICIDS2017 Dataset and the Significant Differences in Detection Performances It Makes*. CRiSIS 2022. **label conflicts**

[3] Arp, D. et al. *Dos and Don'ts of Machine Learning in Computer Security*. USENIX Security 2022. **leakage pitfalls**

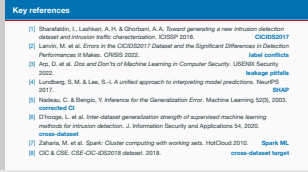
[4] Lundberg, S. M. & Lee, S.-I. *A unified approach to interpreting model predictions*. NeurIPS 2017. **SHAP**

[5] Nadeau, C. & Bengio, Y. *Inference for the Generalization Error*. Machine Learning 52(3), 2003. **corrected CI**

[6] D'hooge, L. et al. *Inter-dataset generalization strength of supervised machine learning methods for intrusion detection*. J. Information Security and Applications 54, 2020. **cross-dataset**

[7] Zaharia, M. et al. *Spark: Cluster computing with working sets*. HotCloud 2010. **Spark ML**

[8] CIC & CSE. *CSE-CIC-IDS2018 dataset*. 2018. **cross-dataset target**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[7] Zaharia, M. et al. *Spark: Cluster computing with working sets*. HotCloud 2010. Spark ML

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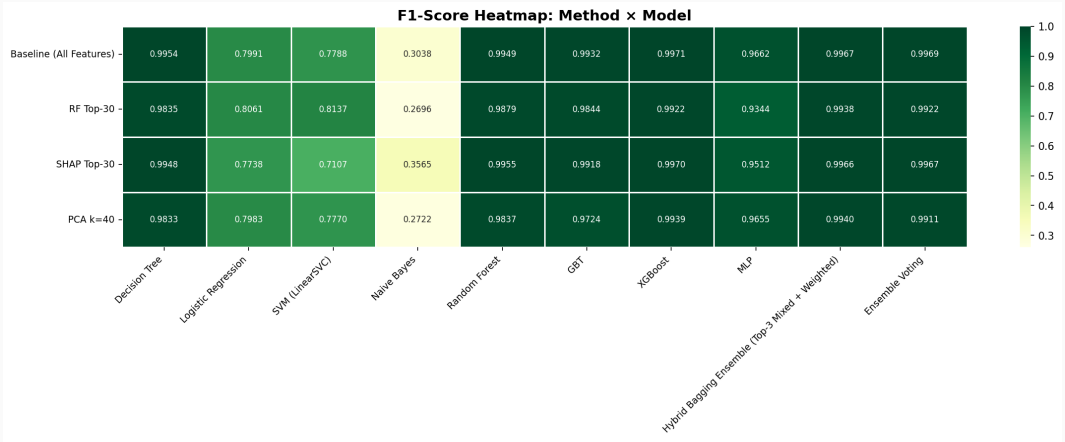
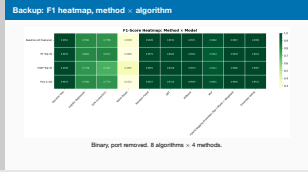
Not presented — shown while taking questions, so sources are on screen if the audience asks. Full list is in the paper.

Backup slides

Backup: F1 heatmap, method × algorithm



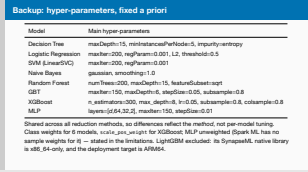
Binary, port removed. 8 algorithms × 4 methods.



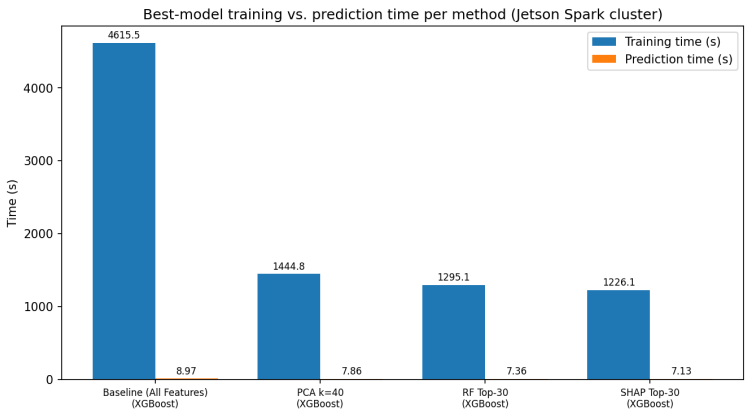
Backup: hyper-parameters, fixed a priori

Model	Main hyper-parameters
Decision Tree	maxDepth=15, minInstancesPerNode=5, impurity=entropy
Logistic Regression	maxIter=200, regParam=0.001, L2, threshold=0.5
SVM (LinearSVC)	maxIter=200, regParam=0.001
Naive Bayes	gaussian, smoothing=1.0
Random Forest	numTrees=200, maxDepth=15, featureSubset=sqrt
GBT	maxIter=150, maxDepth=6, stepSize=0.05, subsample=0.8
XGBoost	n_estimators=300, max_depth=8, lr=0.05, subsample=0.8, colsample=0.8
MLP	layers=[d,64,32,2], maxIter=150, stepSize=0.01

Shared across all reduction methods, so differences reflect the *method*, not per-model tuning. Class weights for 6 models, `scale_pos_weight` for XGBoost; MLP unweighted (Spark ML has no sample weights for it) — stated in the limitations. LightGBM excluded: its SynapseML native library is x86_64-only, and the deployment target is ARM64.



Backup: training and prediction cost



Measured on the Spark training cluster. Latency/throughput *on the edge device* belong to the companion systems paper.

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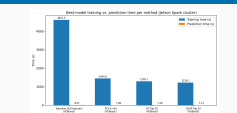

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Backup: threats to validity

- **Internal.** RF/SHAP rankings come from the original training set and stay fixed across resamplings — consistent with the a-priori design, but a slight leak in the ranking step. Re-ranking per resampling is more rigorous, but costly for SHAP. MLP carries the unweighted-training caveat.
- **External.** CICIDS2017 may not reflect current traffic, and benchmark NIDS datasets have documented quality limitations. The cross-dataset conclusion rests on a *stratified sample* of CSE-CIC-IDS2018. Evasive traffic untested. Feature-level dedup itself shifts the benign/attack distribution.
- **Statistical.** $N{=}6$ gives low power: two-sided exact enumeration floors p at $2/2^N$, so $N{=}6$ *just* reaches 0.05. Resamplings overlap, violating t-independence — hence Nadeau–Bengio as the primary basis. Nested CV with larger N is the next reinforcement.

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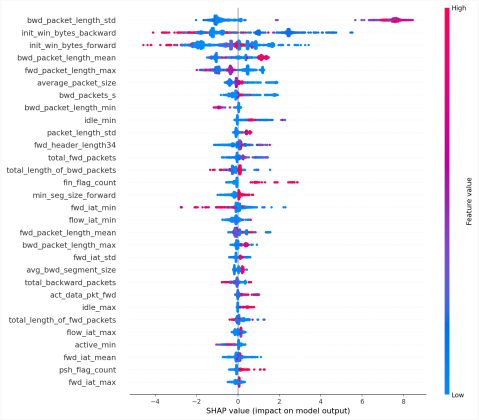
Backup: SHAP feature attribution



SHAP is computed on a *class-stratified* sample so both benign and attack traffic are represented in the ranking.